13. Kamalam went to play a lucky draw contest. 135 tickets of the lucky draw were sold. If the probability of Kamalam winning is $\frac{1}{9}$, then the number of tickets bought by Kamalam is

(A) 5 (B) 10 (C) 15 (D) 20

14. If a letter is chosen at random from the English alphabets $\{a, b, \ldots, z\}$, then the probability that the letter chosen precedes $x$

(A) $\frac{12}{13}$ (B) $\frac{1}{13}$ (C) $\frac{23}{26}$ (D) $\frac{3}{26}$

15. A purse contains 10 notes of ₹2000, 15 notes of ₹500, and 25 notes of ₹200. One note is drawn at random. What is the probability that the note is either a ₹500 note or ₹200 note?

(A) $\frac{1}{5}$ (B) $\frac{3}{10}$ (C) $\frac{2}{3}$ (D) $\frac{4}{5}$

## Unit Exercise - 8



1. The mean of the following frequency distribution is 62.8 and the sum of all frequencies is 50. Compute the missing frequencies $f_1$ and $f_2$.

| Class Interval | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 | 100-120 |
|---|---|---|---|---|---|---|
| Frequency | 5 | $f_1$ | 10 | $f_2$ | 7 | 8 |

2. The diameter of circles (in mm) drawn in a design are given below.

| Diameters | 33-36 | 37-40 | 41-44 | 45-48 | 49-52 |
|---|---|---|---|---|---|
| Number of circles | 15 | 17 | 21 | 22 | 25 |

Calculate the standard deviation.

3. The frequency distribution is given below.

| $x$ | $k$ | $2k$ | $3k$ | $4k$ | $5k$ | $6k$ |
|---|---|---|---|---|---|---|
| $f$ | 2 | 1 | 1 | 1 | 1 | 1 |

In the table, $k$ is a positive integer, has a varience of 160. Determine the value of $k$.

4. The standard deviation of some temperature data in degree celsius (°C) is 5. If the data were converted into degree Farenheit (°F) then what is the variance?

5. If for a distribution, $\sum(x-5)=3$, $\sum(x-5)^2=43$, and total number of observations is 18, find the mean and standard deviation.

6. Prices of peanut packets in various places of two cities are given below. In which city, prices were more stable?

| Prices in city A | 20 | 22 | 19 | 23 | 16 |
|---|---|---|---|---|---|
| Prices in city B | 10 | 20 | 18 | 12 | 15 |

7. If the range and coefficient of range of the data are 20 and 0.2 respectively, then find the largest and smallest values of the data.



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